

Strategic Practice 3

- 1.1) M_2 : event Monty opens door 2
 M_3 : event Monty opens door 3

If Monty is choosing b/w opening doors 2 & 3
 he chooses door 2 w/ probability p , $\frac{1}{2} \leq p \leq 1$

- 2) C_i : event car is behind door i $i \in \{1, 2, 3\}$

What is probability that the strategy of ALWAYS
 switching doors succeeds?

$$P(\text{get car}) = P(\text{get car} | C_1) P(C_1) \\
+ P(\text{get car} | C_2) P(C_2) \\
+ P(\text{get car} | C_3) P(C_3)$$

WLOG, assume contestant picked
 door 1 originally

$$= 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3}$$

- b) S : event that switching strategy succeeds
 D_i : event that door i is initial pick $i \in \{1, 2, 3\}$
 WLOG, assume car is behind door 1.

$$P(S | M_2) = \frac{0}{P(S | M_2, D_1) P(D_1 | M_2)} \\
+ \frac{P(S | M_2, D_2) P(D_2 | M_2)}{0} \\
+ P(S | M_2, D_3) P(D_3 | M_2)$$

$$= P(S | M_2, D_3) P(D_3 | M_2) \\
= 1 \cdot P(D_3 | M_2)$$

$$= \frac{P(M_2 | D_3) P(D_3)}{P(M_2)}$$

$$\left(\begin{array}{l} P \\ P(M_2) = \frac{P(M_2 | D_1) P(D_1)}{0 \dots 0} \end{array} \right) \frac{1}{3}$$

$$\begin{aligned}
 & \begin{array}{l} + \cancel{P(M_2|D_2)} \cancel{P(D_2)} \\ + \cancel{P(M_2|D_3)} \cancel{P(D_3)} \end{array} = \frac{1+p}{3} \\
 & = \frac{1 \cdot \frac{1}{3}}{(1+p) \cdot \frac{1}{3}} = \frac{1}{1+p}
 \end{aligned}$$

$$\begin{aligned}
 c) \quad P(S|M_3) &= \cancel{P(S|M_3, D_1)} P(M_3|D_1) \\
 &+ P(S|M_3, D_2) P(M_2|D_2) \\
 &+ \cancel{P(S|M_3, D_3)} \cancel{P(M_3|D_3)} = 0 \\
 &= P(D_2|M_3) \\
 &= \frac{P(M_3|D_2) P(D_2)}{P(M_3)}
 \end{aligned}$$

$$\begin{array}{l}
 P(M_3) = \cancel{P(M_3|D_1)} P(D_1) \\
 + \cancel{P(M_3|D_2)} P(D_2) \\
 + \cancel{P(M_3|D_3)} P(D_3) = \frac{1}{3} \cdot (2-p)
 \end{array}$$

$$= \frac{\frac{1}{3}}{\frac{1}{3}(2-p)} = \frac{1}{2-p}$$

1.2) X, Y, Z are discrete r.v.s

a) FALSE

counterexample:

X : weather outside

Y : flipping a coin

Z : getting 12e coin

b) FALSE

counterexample

X : flipping a H on first coin flip

Y : flipping a H on second coin flip

Z : getting 2 different outcomes on first and second coin flips

c) FALSE

Counterexample:

There are 2 opponents you can play and you cannot distinguish b/w them. You're randomly assigned to play one. One player you are evenly matched against. The other one is better than you.

X: outcome playing evenly matched player

Y: outcome playing more advanced player

Z: no games have been played yet

d) TRUE

$$P(X=a|Z=z) = P(Y=a|Z=z)$$

$$\rightarrow \text{then } P(X=a) = P(X=a|Z=z) P(Z=z)$$

$$P(Y=a) = P(Y=a|Z=z) P(Z=z)$$

$$\therefore P(X=a) = P(Y=a)$$

2.1) events A, B, E

a) given $P(A|E) < P(B|E)$

and $P(A|E^c) < P(B|E^c)$

then $P(A) > P(B)$ is NOT possible

$$P(A) = P(A|E)P(E) + P(A|E^c)P(E^c)$$

$$P(B) = P(B|E)P(E) + P(B|E^c)P(E^c)$$

$$\therefore P(A) < P(B)$$

b) given $P(A|B, E) < P(A|B^c, E)$

and $P(A|B, E^c) < P(A|B^c, E^c)$

then $P(A|B) > P(A|B^c)$ is possible

example from book

A: surgery succeeds

B: doctor 1 or 2

C: surgery 1 or 2

	Doc 1		Doc 2	
	success	failure	success	failure
81	20	20	2	8
82	10	0	81	9
	80	20	83	17

2.2) a) H: stampy (elephant) gets hurt

D: is dealer

I: has a lot of ivory

b) Lisa says $L(D|I) > \cancel{L(D|I^c)}$ $L(I)$
 $L(H|I) > L(H|I^c)$ implicitly
Homer says $L(H|I) < L(H|I^c)$

c) Even if $P(H|I) < P(H|I^c)$, Homer does not condition on the fact that Blackheart is a dealer

Homer does not realize that observing Blackheart has so much ivory makes it more likely Blackheart is a dealer, which makes it more likely that he'll hurt Stampy.

3.1) start w/ \$1M

each game $\frac{1}{3}$ chance of winning \$1
 $\frac{2}{3}$ " " losing \$1

$$P(\text{up by } \$2) = P(\text{winning all } \$1,000,002)$$

$$= \frac{1 - \left(\frac{2/3}{1/3}\right)^{1,000,000}}{1 - \left(\frac{2/3}{1/3}\right)^{1,000,002}}$$

$$= \frac{2^{1,000,000} - 1}{2^{1,000,002} - 1} < \frac{1}{4}$$

$$\text{b/c } 2^{1,000,002} - 4 < 2^{1,000,002} - 1$$

4.1) a) $P(A \text{ wins series})$

$$= \binom{4}{4} p^4 + \binom{4}{3} p^4 (1-p) + \binom{5}{3} p^4 (1-p)^2 + \binom{6}{3} p^4 (1-p)^3$$

b) answer (a) does not depend on whether the teams play all 7 games or stop playing as soon as one team has won 4.

taking the later approach makes no calculations or assumptions on what the outcome of the remaining games could be - all possibilities are still counted, which is the all 7 games approach

mathematical verification

$$\binom{4}{4} p^4 + \binom{4}{3} p^4 (1-p) + \binom{5}{3} p^4 (1-p)^2 + \binom{6}{3} p^4 (1-p)^3$$

$$= 35 p^4 - 84 p^5 + 70 p^6 - 20 p^7$$

$$= \binom{7}{4} p^4 (1-p)^3 + \binom{7}{3} p^5 (1-p)^2 + \binom{7}{6} p^6 (1-p) + \binom{7}{7} p^7$$

4.2) n independent experiments

$$P(\text{success}) = p$$

$$P(\text{failure}) = 1-p$$

show that conditional on # of successes, the possibilities of the list of outcomes for each

experiment is equally likely.

$$i \text{ of successes} = i \quad 0 \leq i \leq n$$

Given i successes, 2 possibility for the outcomes of n experiments can be made by scrambling i S events and $n-i$ F events

success failure

For any possibility of the n experiment outcomes
the probability of that possibility is

$$\binom{n}{i} p^i (1-p)^{n-i}$$

The solution uses n.v.s!

Let X_j be 1 if j th experiment is a success and 0 otherwise. Let $X = X_1 + \dots + X_n$ be total # of successes. For any k and any $z_1, \dots, z_n \in \{0, 1\}$ with $z_1 + \dots + z_n = k$:

$$P(X_1 = a_1, \dots, X_n = a_n \mid X = k) = \frac{P(X_1 = a_1, \dots, X_n = a_n, X = k)}{P(X = k)}$$

$$= \frac{p^k (1-p)^{n-k}}{\binom{n}{k} p^k (1-p)^{n-k}}$$

$$= \frac{1}{\binom{n}{k}}$$

all $\binom{n}{k}$ possible sequences have equal probability of occurring given $X=1$

4.3) a) $X \sim \text{Bin}(n, p)$ $Y \sim \text{Bin}(m, p)$ X and Y are independent

X is the number of successes of n Bernoulli trials
 Y " " " " " " " " " "

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$X+Y$ " " " " " " " " $n+m$ " "
 so $X+Y \sim \text{Bin}(n+m, p)$
 needs to mention each Bernoulli trial has p probability of success

b) $P(X-Y = k) = \sum_{j=0}^n P(X-Y = k | X=j) P(X=j)$ LOTP
 $\uparrow$
 when $j \geq 0$, $Y = -k$
 which is impossible since the number of successes of n Bernoulli trials can't be negative.

The solution simply states: A binomial can't be negative, but $X-Y$ is negative with a positive probability

c) $P(X=k | X+Y=j)$
 $= \frac{P(X+Y=j | X=k) P(X=k)}{P(X+Y=j)}$
 $= \frac{P(X+Y=j | X=k) \binom{n}{k} p^k (1-p)^{n-k}}{\binom{n+m}{j} p^j (1-p)^{n+m-j}}$
 $= \frac{P(Y=j-k)}{\text{" "}}$
 $= \frac{\binom{m}{j-k} p^{j-k} (1-p)^{m-j+k} \binom{n}{k} p^k (1-p)^{n-k}}{\binom{n+m}{j} p^j (1-p)^{n+m-j}}$
 $= \frac{\binom{m}{j-k} \binom{n}{k}}{\binom{n+m}{j}}$ Hypergeometric distribution